HYPER-3

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Author Note

The HYPER-3 study evaluates which of three daily doses should go into phase III and whether any of the doses harms the people taking it. The target effect of dose on systolic blood pressure is identified via the backdoor route. The trial question is answered by the intent-to-treat estimate of -1.53 (90% WALD -2.97 to -0.10), for which no decision threshold was recorded. The analysis yields a realized treatment effect of -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). Across the remaining evaluations, the estimated effect on survivors is 7.09 (90% WALD 4.84 to 9.33), and the final estimate is 6.20 (90% WALD 3.98 to 8.42).

Abstract

HYPER-3 studied three daily doses of a new antihypertensive in adults who already have high blood pressure or are on their way to it, watching them every week for six months. The evaluation asked which of the three daily doses should go into phase III and whether any of them was harming the people taking it. The effect of dose on sbp is identified via the backdoor route. Randomization serves as the strongest standing assumption. The intent-to-tert estimate is answered by the intent-to-treat estimate. The intent-to-treat estimate is reported as -1.53 with a 90% WALD interval from -2.97 to -0.10, while the estimated effect on survivors, the realized treatment effect, and the final estimate are reported without a recorded decision threshold.

Reported quantities: intent-to-treat estimate; Estimated effect on survivors; Realized treatment effect; Final estimate

Introduction

HYPER-3 asks which of three daily doses should go into phase III, and whether any of them is harming the people taking it. This target is identified and is answered by the intent-to-treat estimate. No decision threshold was recorded for this quantity. An interval lying entirely to one side of a decision value settles whether an effect satisfies that criterion, whereas an interval spanning the value leaves the question unsettled. Because no threshold was specified, this report provides the intent-to-treat estimates and their intervals without assessing them against a particular benchmark.

Methods

To resolve the clinical decision established in the introduction, the target of the analysis is the effect of assigned dose on systolic blood pressure, identified via the backdoor route. The trial follows adults with elevated blood pressure weekly for six months across four study arms that compare three daily doses against standard care. Identification rests on randomized assignment, which licenses the effect of assigned dose on the outcome in the randomized population without conditioning on downstream adherence, dropout, or pressure readings. The study enrolls 400 units across four arms to evaluate effects inside pre-specified age strata, exceeding the 71 units needed for a pooled analysis. The design parameters are recorded in Table 1. Rather than estimating four isolated arm means, the analysis fits a continuous dose-response curve to inform dose selection for phase III testing. The estimation procedure specifies a saturating kernel per treatment in a surface specification, builds a single model specification evaluated by the likelihood and optimizer, and fits the surface. A sequential design establishes boundaries to stop any dose arm that worsens blood pressure relative to standard care, evaluated overall and within each age stratum. Sequential inspections occur every second calendar week beginning at week 12, when three eighths of the safety endpoints are available. The monitoring procedure conducts eight safety reviews across the trial, evaluating twelve contrasts against the sequential boundaries at each interim check.

Results

The analysis addresses the comparison specified in the introduction under the framework set out in the methods. The intent-to-treat estimate is -1.53 (90% WALD -2.97 to -0.10). The estimated effect on survivors is 7.09 (90% WALD 4.84 to 9.33). The realized treatment effect is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). The final estimate is 6.20 (90% WALD 3.98 to 8.42). Table 2 reports these estimated quantities with their intervals and thresholds. Figure 1 displays the estimated quantities with their intervals, where the dashed rule marks the decision threshold and a row whose interval crosses it is unsettled rather than null.

INTENT-TO-TREAT ESTIMATE	ESTIMATED EFFECT ON SURVIVORS	REALIZED TREATMENT EFFECT	FINAL ESTIMATE
-1.53	7.09	-4.80 mmHg	6.20
[-3.0, -0.1] (90% WALD)	[4.8, 9.3] (90% WALD)	[-6.2, -3.4] (90% WALD)	[4.0, 8.4] (90% WALD)

What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

From 0004-case-study-hypertension.md: # 0004 — HYPER-3: a sequential-trial case study, and the module it exposed **Kind**: progress + decision. **Opened**: 2026-08-21. **Status**: landed on `develop`; the new public API is destined for 1.1.0. ## Why Nothing in `axiom` was wrong, but nothing in `axiom` showed a reader what the pieces are *for* when they are all pointed at one decision. The notebook series under `nbs/<subpackage>/` demonstrate every public symbol (gate 12) and the `nbs/end-to-end/` series walks each pillar, but neither answers "what does using this look like on a real problem, start to finish, with the awkward parts

From 0005-basis-response-families.md: # 0005 — Basis response families: polynomial, spline, piecewise linear **Kind**: decision. **Opened**: 2026-08-21. **Status**: landed on `develop`; part of the 1.1.0 surface. ## Why Note [0004](0004-case-study-hypertension.md) §0004.3 recorded a gap the HYPER-3 case study walked into: every response kernel `axiom.surface` shipped is **monotone and saturating**, and the case study's oldest age stratum has a dose-response that *turns over*. The Hill fit could not describe it, and failed informatively — an arm-mean residual of 8 mmHg at the top dose against ≤1.5 in the strata where the family fit

From README.md: # HYPER-3 — a sequential dose-finding trial in hypertension A worked case study that runs the whole of `axiom` against one question, end to end. Every number in the six notebooks is computed; the synthetic world they share lives in [`hyper3.py`](hyper3.py) beside them. ## The question > *We have a new antihypertensive. We want to know which of three daily doses to take > into phase III, in adults who already have high blood pressure or are on their way to > it. We are going to watch these people every week for six months. If one of the doses > is making people worse than the care they would ha

Model checking

No diagnostic checks of the model fit were recorded for this analysis. Model checking can lower confidence in an estimate, but it cannot warrant an estimate on its own. Because no diagnostics were performed, potential misspecifications in the fit remain unexamined. The absence of checks leaves the empirical behavior of the specification untested rather than sound.

Discussion

Because no decision threshold was recorded for the intent-to-treat estimate, the estimated effect on survivors, the realized treatment effect, or the final estimate, this report presents the quantities without judging dose selection or potential harm. Identification via the backdoor route licenses reading these estimates as effects of the treatment rather than unadjusted associations, strictly under the assumptions outlined in the methods. Readers should consult the limitations section regarding the scope and plausibility of those standing assumptions. Resolving the choice of dose for phase III now requires setting explicit decision thresholds against which the estimates recorded in the results can be evaluated.

Conclusions

The question of which of three daily doses should advance to phase III and whether any dose causes harm yields an intent-to-treat estimate of -1.53 (90% WALD -2.97 to -0.10). Because no decision threshold was recorded, this estimate is reported without a decision judgment. For the same reason, the estimated effect on survivors, the realized treatment effect, and the final estimate are reported and not judged. These answers rest on the identified design and the assumptions specified in the methods.

Limitations

The effect is identified under the structural assumptions established in the methods, leaving no assumption unresolved in the present record. This identification licenses the interpretations advanced earlier, claiming that the modeled relationships capture the underlying process. The analysis would be challenged by unobserved confounding or structural misspecifications that violate these baseline conditions. Direct empirical tests of these conditions across alternate settings would resolve doubts about whether the identification holds outside the current data.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
intent-to-treat estimate	02_itt	-1.53 (90% WALD -2.97 to -0.10)	02-what-it-identifies.ipynb · LinearEstimate
Estimated effect on survivors	02_survivors	7.09 (90% WALD 4.84 to 9.33)	02-what-it-identifies.ipynb · LinearEstimate
Realized treatment effect	03_realized	-4.80 mmHg (90% WALD -6.17 to -3.44 mmHg)	03-sizing-it.ipynb · LinearEstimate
Final estimate	06_final_estimate	6.20 (90% WALD 3.98 to 8.42)	06-running-the-trial.ipynb · LinearEstimate

Table 1. Quantities and their sources

Field	Value
cells	92
figures	9
gaps_closed	1
notebooks	6
evidence hash	dfab0ed3139b8193200340fb3d4aa91e489b1df8aae666c7affe3ed3029689ab
narration: abstract	gemini-3.5-flash-lite — verified
narration: title_page	gemini-3.7-flash — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified

narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 2. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	route	backdoor
Identification	status	identified
HYPER-3, 1 — The trial, and the synthetic world it runs in	cells	14
HYPER-3, 1 — The trial, and the synthetic world it runs in	sections	12
HYPER-3, 2 — What the trial identifies, and what it does not	cells	12
HYPER-3, 2 — What the trial identifies, and what it does not	sections	9
HYPER-3, 3 — How big, allocated how, powered for what	cells	14
HYPER-3, 3 — How big, allocated how, powered for what	sections	9
HYPER-3, 4 — The dose–response curve, and which doses phase III should test	cells	15
HYPER-3, 4 — The dose–response curve, and which doses phase III should test	sections	10
HYPER-3, 5 — The sequential design: boundaries, and what they cost	cells	19
HYPER-3, 5 — The sequential design: boundaries, and what they cost	sections	13
HYPER-3, 6 — Running it: the interim that stops an arm, and the readout	cells	18

HYPER-3, 6 — Running it: the interim that stops an arm, and the readout	sections	14

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
intent-to-treat estimate	-1.53	—	-2.97 to -0.10 (90% WALD)	—	—
Estimated effect on survivors	7.09	—	4.84 to 9.33 (90% WALD)	—	—
Realized treatment effect	-4.80	mmHg	-6.17 to -3.44 (90% WALD)	—	—
Final estimate	6.20	—	3.98 to 8.42 (90% WALD)	—	—

Figures

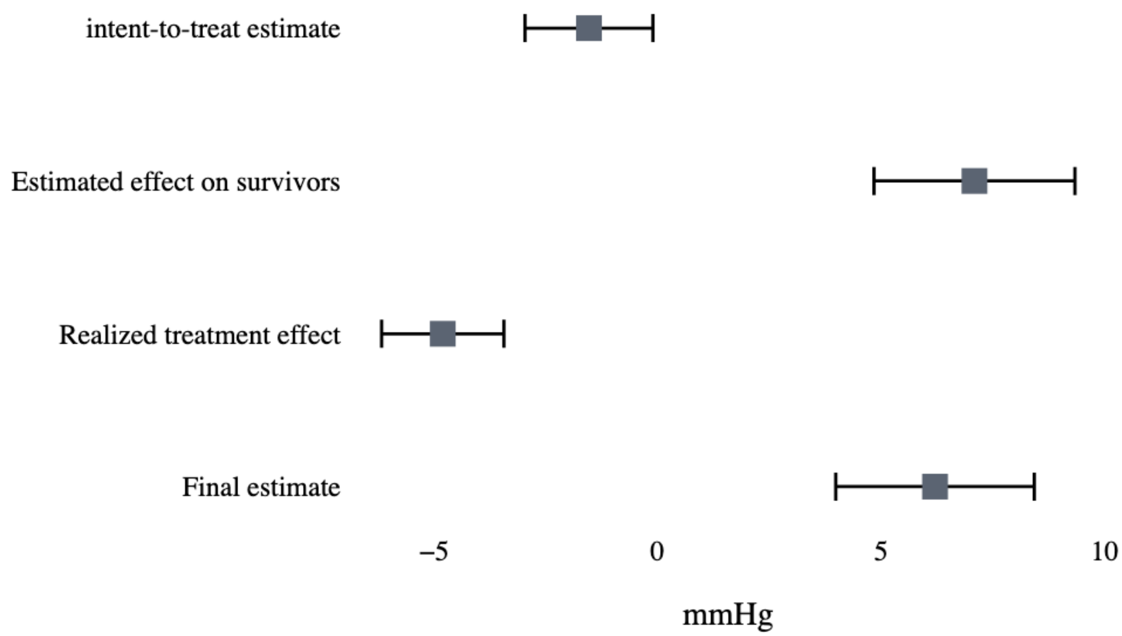

Figure 1. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.